KVRouite – Documentation

supported by kinomap

***KVRouite** stands for **"Kinomap Video Route Suite"** — an open-source tool to synchronize videos with GPX/FIT data (officially supported by Kinomap).*

Introduction

KVRouite aligns your video with GPX/FIT data so each frame corresponds precisely to its position on the map, making uploads to Kinomap seamless.

When the video and GPX/FIT file are imported the application asks if the video start is already synchronized with gpx start. If it's not the case it is advised to start by finding a matching gpx and video location. It can be anywhere in the video, not especially at the start. Until this synchronization is not done, video and gpx edit and playback are not synchronized.

Modes

The program can be used in three different modes:

GPX Mode

This is the default setting, where a pre-cut video (edited outside of KVRouite) and its corresponding GPX file are loaded. In this mode, only the GPX file is modified.

- The start of the GPX can be synchronized, and pauses can be cut.
- Various manipulations of individual points or sections are possible, such as modifying time, altitude, or gradient.
- Errors in waypoints or timestamps are detected upon loading and marked in the diagram. These can be removed via the "More (...)" menu.
- Multiple GPX files can be loaded and appended with a 1-second gap (e.g., multiple GoPro GPX files).
- It is also possible to edit a GPX file without a video if needed, for example, to merge multiple GoPro files. However, GoPro files should be recorded with a 1Hz resolution.
- There are various tools available to extract data from GoPro metadata and prepare it accordingly, such as:
  https://goprotelemetryextractor.com/free/#
  In this tool, select:
    - Frequency: 1Hz
    - Altitude: corrected
    - Save as: GPX or Virb

Video Mode

If you only need to merge multiple video files (e.g., from a GoPro) and optionally cut them, you can select “Edit Video” from the menu.

> **Copy Mode:**
> The video is not re-encoded,
> but instead handled by LossLessCut – meaning it is simply *copied*.
> This may result in slightly abrupt transitions at cut points. However, these are negligible compared to the overall uncut video or, for example, waiting at a traffic light.
>
> **Encoder Mode:**
> The video is re-encoded using the settings from the Encoder Setup!
> Options configured in the Encoder Setup, such as GPU usage, will be applied.
> Cuts will also be transitioned smoothly using a crossfade (xfade) effect.
> Copy Mode needs ffmpeg and ffprobe on your computer. They are not part of KVRouite. If they are missing, Copy Mode cannot be selected and Encode Mode is used instead.

When you open a video, you are prompted to select a mode. You can change it later.

AutoCut Video + GPX Mode

This setting, also found in the "Config" menu, trims the video and simultaneously removes the corresponding sections from the GPX file (traffic lights, breaks, etc.). The video is then already synchronized.

Edit view

This mode is convenient if you edit only the gpx file or your gpx files are longer than video.

To clarify the layout:

- At the top, as usual, there is the menu bar with options like File, Playlist, Config, etc., and their submenus.
- The window is divided into four areas: two above the timeline, two below it. Which module sits in which area is up to you - map, chart, Chart Flow or GPX table. Move the mouse into an area and a small handle appears at its top edge; a right-click anywhere in the area opens the same list. Picking a module that currently sits somewhere else makes the two areas swap. The video can only be in the upper row and changes sides there.
- The player buttons sit directly under the video picture, the GPX button bar under the GPX table - or, if you prefer, over the table, under or over the map, or floating with a grip anywhere in the window (Config > GPX Bar Position). Each module carries its own controls, so they travel with it when you move it to another window.
- Across the middle, over the full width of the window, runs the timeline.
- In the map, the map control buttons are at the top right, and the switch for toggling between satellite view and terrain view is at the bottom left.

- In the list, the time associated to gpx points can be edited by double click. This changes only a single point. If you need to shift all next points, see “chT” button (explained later).

Menu File

New Project
Start a new Project, all loaded files and edits will be deleted and you start from new.

Load Project
Load an already saved Project, all files and edits will be reloaded

Open Recent
Choose one of the last loaded projects/files

Import GPX/FIT
Attention: GPX/FIT Files will be loaded in Slot 1 (read Slot description)

Loads a GPX or a FIT file.
It will then be displayed on the map, in the diagram, the mini-diagram, and the GPX list.
If a file is already loaded and "Open GPX" is selected again, you will be given the choice to append it or load it as a new file.
If you choose to append it (which is useful for GoPro GPX files), the new file will be added to the previous one with a 1-second gap. This makes it easy to merge multiple GPX files of a route seamlessly.
If you choose "New," the old GPX file will be removed from the tool, and the new one will be loaded.

Import Video

Loads one or more video files.
If multiple videos are loaded, they will be played sequentially in the player, and a blue marker will be placed between them in the timeline to indicate separation visually.
The total duration of all loaded videos is displayed at the bottom left of the player, while the current video time is shown in the top right during playback.

Gopro-Extractor
Attention: Gopro – GPX will be loaded in Slot 2 (read Slot description=

Several files selected at once go into the playlist in recording order, not in the order they were clicked: the recording time is read from the video container (the value ffprobe shows as creation_time), and GoPro chapters of one recording - GX010039, GX020039, GX030039 - follow each other by chapter number. Where the container has no time, the file's creation time is used. The order is printed to the console. Sorting by name would be wrong: GX020039 would come after GX010042.

Experimental Feature - This feature is currently in testing and may not work with all GoPro models or firmware versions.

Requirements:
- GoPro videos with embedded GPS data (typically Hero 5 and newer)
- Videos must contain GPS signals (recorded outdoors with GPS reception)

Usage
1. Load your GoPro videos via **File → Import Video**
2. Select **File → Import Video → Extract Gopo-GPS**
3. Read the experimental warning and click **Yes** to continue

4. Choose Files for extraction:
  - All - will be load all files
  - Selected – only selected files will be loaded
  - Ony First – only the first will be loaded (for sync Slot1 ypu need ony the first one)
5. The extraction process will start automatically for all loaded videos
6. After completion, you'll see an elevation data quality notice

Important Notes
- Raw GoPro elevation data is often inaccurate due to barometric sensor limitations
- For accurate elevation data:
  - Use **Update Elevation** in the GPX Control panel (requires Mapbox key)
  - Apply **smoothing** with the smoothing tool
- If GPX data is already loaded, you'll be prompted to clear it before extraction
- Only works with GoPro files containing valid GPS metadata

Troubleshooting
- If extraction fails, ensure your GoPro has GPS reception during recording
- Check that the video files contain GPS metadata (use GoPro Player to verify)
- Some GoPro models/firmware may not be fully supported

Safe Project

safe the actual loaded files and edits in a project file, you can work later on it, but you cant undo any edits!

Export GPX

Saves the GPX file with all applied changes.

Export Video

Saves the video with all cuts applied using the LossLessCut method (Copy-Mode) or the Encoder method (Encode-Mode)

Hint: You choose Copy-Mode or Encode-Mode in Menu “Config” -> “Edit Video”
Copy-Mode is only offered when ffmpeg and ffprobe are available on your computer.

Stopping a running export

In Encode-Mode the encoder window shows the progress. Close on that window asks "Stop the export?": Yes stops the encoder within a second or two, deletes the incomplete output file and closes the window; No leaves it running. Closing the main window while an export runs asks the same question. In Copy-Mode the export window has a Cancel button that stops ffmpeg.

Menu Edit

In this Menu you get the Undo action and the Undo history.

Undo – Shortcut: STRG +Z
Reverts the last action. You can always choose STRG+Z

Undo history…
Every undo step carries a name and a time - for example "Cut 1952.118-1954.118 s (video + GPX)", "chT rows 1890..1893 (00:32:30.011 - 00:32:33.011)", "Smooth (whole track)". Edit > Undo history lists them, newest first; the top entry is what Ctrl+Z undoes next. Select an entry and press "Undo to here": that step and every step above it are undone, after a confirmation that says how many. A single step cannot be taken out of the middle - the list holds states, not actions. Taking back one cut on its own is what the right-click on a cut in the timeline offers.

Menu Playlist

Displays the loaded videos.
Videos can be removed by clicking on them,
The "Reorder" menu item opens a pop-up window where you can change the order of the videos if you loaded them incorrectly. But be careful: Don't use this menu item if you already have synced, edited, or overlays!

Menu View

Window headers

Each of the four windows can show a small header with the name of the module in it. The header costs about 20 pixels of height, so it is off by default - the switch handle and the right-click menu do the same job without taking up room.

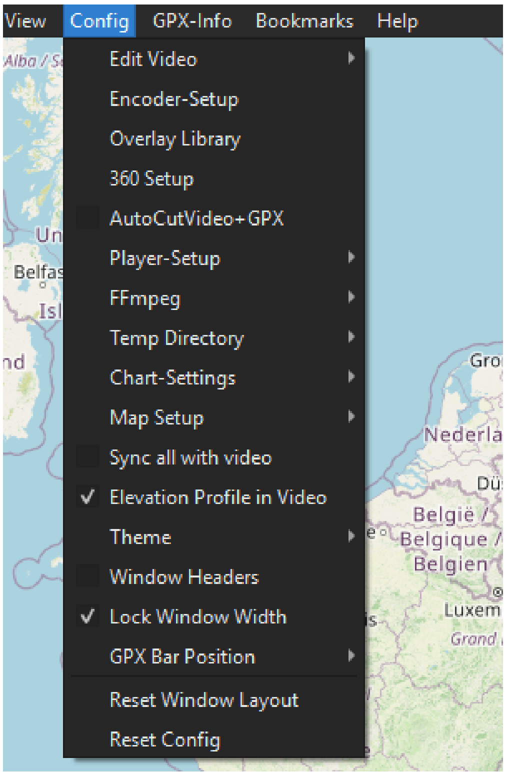
Elevation profile in the video

Shows the Chart Flow over the video picture, bottom left, dark and translucent. Drag it anywhere inside the picture by the small handle in its top left corner; the position is remembered. It stays inside the picture, not on the black bars that appear when the video does not fill the window.

360° Video – Shortcut: V
Turns the 360° view on and off. KVRouite switches it on by itself when the material is equirectangular, that is twice as wide as it is high.
The picture you see is a real projection out of the sphere, not a crop: straight lines stay straight, and you can pan across the seam without an edge.
Drag in the picture with the left mouse button to look around. Turn the mouse wheel to zoom. CTRL + arrow keys do the same in steps, CTRL + / CTRL - zoom, CTRL 0 resets the view.
Each video keeps its own viewing direction, and it is stored in the project file. View → Apply 360° view to all videos copies the current one to every video, which is what you want for

material from one camera.
The view you pick is what gets rendered. Preview and export build the same timeline, so the exported video is a normal 16:9 video showing exactly the section you chose.
Keyframes for the viewing direction: the view can change over time. Press the KF button (shown in 360° mode only) to store the current view at the marker as a keyframe. Between two keyframes the camera pans smoothly; before the first keyframe it pans from the view the video starts with, after the last keyframe the view holds. Set the keyframe first and then drag the picture: the keyframe under the marker takes the new view. Dragging anywhere else only previews the view until you play, jump or press KF. Keyframes appear as green diamonds at the top of the timeline – see chapter Timeline for moving, removing and hard cuts. Every keyframe action can be undone with CTRL+Z.

Menu Config



Edit Video

If you want to cut the video, you need to activate one of the two modes (see below). This will enable additional buttons in the video control section.
On first activation, you'll be asked whether to index keyframes or not.
Indexing allows you to cut *exactly* at keyframe positions. While this is not mandatory, it can help

produce cleaner cuts.
Once editing is activated, you'll see the overlay Edit: On in the video window.

Edit Video / Copy Mode:
As mentioned earlier, this uses LossLessCut.
The advantage: The video is not re-encoded – it is built using the copy function only.
The downside: Transitions at cut points may be abrupt or hard cuts.

Edit Video / Encode Mode:
The entire video is re-encoded using the settings from the Encoder Setup.
Crossfades (xfades) – smooth transitions between clips – are applied.
You can also choose the video resolution, quality, and frame rate (FPS) during encoding.
⚠️ **Important:**
Cuts and overlays must be at least 2 seconds apart – ideally 4 seconds.
This ensures proper transitions and prevents rendering issues.

Example: Cut 1: from second 10 to 22 - Overlay start: at second 24

Encoder Setup *(Only active in Encoder Mode)*
This is where you configure the encoder:

- **Resolution**: Sets the output video resolution (16:9)
- **Container**: e.g., x265 (*Recommended: x265 – fewer issues with crossfades*)
- **Hardware**: Select the hardware (e.g., Nvidia or CPU) to be used for encoding
- **Quality (CFR)**: The lower the value, the higher the quality
  (Tip: the value of 20 results in almost no visible quality loss)
- **Preset**: Controls compression and encoding speed – the faster the preset, the larger the file
- **FPS**: Frames per second
- **XFade: e.g., 2s – sets the duration of the crossfade between clips**
- **Detect HW**: Automatically detects your encoding-capable hardware.
  After detection, only the supported hardware will be selectable – not "everything".

  Tip: Try encoding a short 1-minute video with different settings to see what works best for you.

Presets
The row at the top of the Encoder Setup. Choosing a saved preset fills in the fields below; OK makes them the active settings, as with any other change in this window. "Save…" stores the fields as they stand under a name (an existing name is replaced after a question), "Delete" removes the chosen preset and leaves the fields as they are. The dialog opens with the preset that matches the fields exactly, otherwise "(none)". A preset holds resolution, container, hardware, CRF, preset, bitrate, FPS and X-Fade.

A second tab: Audio
Since 7.0 the Encoder Setup has a second tab, **Audio**, for the sound track, the vehicle damper and the voice remover. It is described in the chapter **Sound**.

Overlay Setup

First of all: What are overlays?
An overlay is an image that is displayed *on top* of the video at a specific time and for a set duration. For example, you can show a short logo at the beginning of the video, or a quick intro screen. Recommended: Use .jpg images for overlays.

Within the menu, you can preconfigure up to three overlays, so you don't have to reconfigure them every time.
This makes it easy to automatically show your logo at the beginning and/or end of a video.

You need to set the image path, and optionally scale it if the image is very large (e.g., choose 0.5 or 0.1 to reduce its size).
You can also choose the corner in which the overlay should appear, and use dx and dy to shift it slightly inward from the edge, so it doesn't appear "crammed" into the corner.

How to insert an overlay:
Use the Ovl button in the Video Editor Control panel.
When pressed, it inserts the overlay at the currently selected point in the timeline.

In the menu, you can:

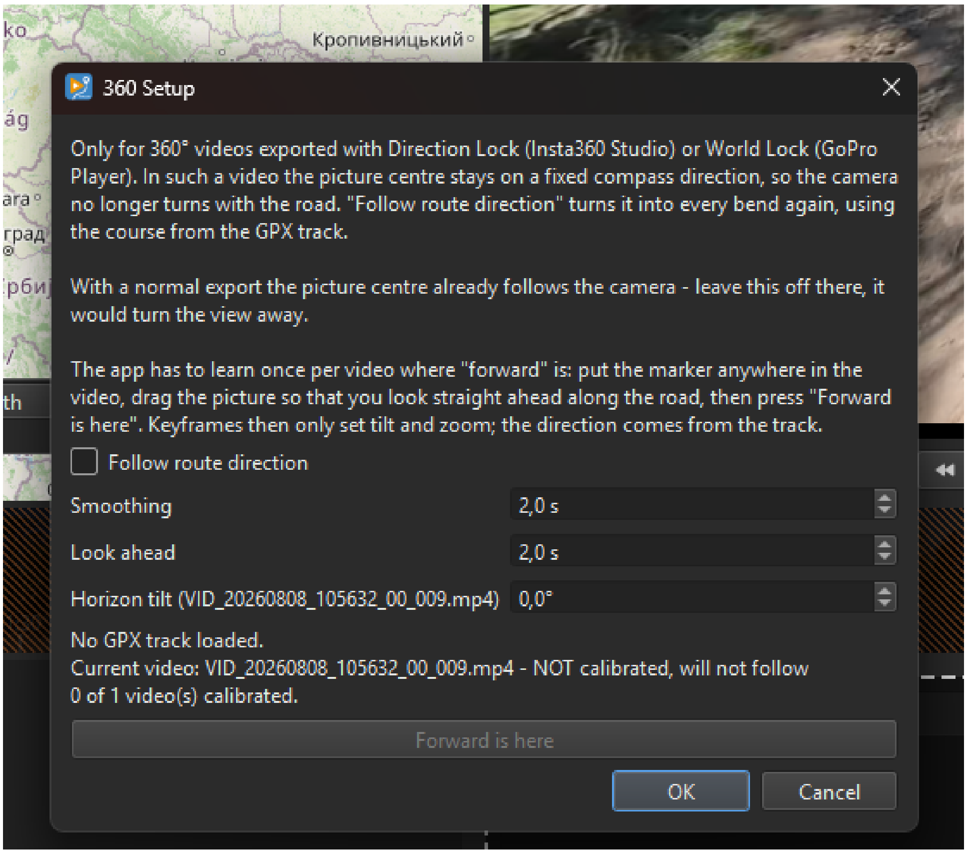
- Choose one of the predefined overlays
- Load a custom overlay ad hoc
- Set display duration
- Optionally set fade-in and fade-out times (if the overlay supports transparency)

⚠️ **Important:** Overlays can be shown for a maximum of 30 seconds.
Note: Platforms like Kinomap generally discourage the use of overlays – so it's best to limit them to a brief logo only.

360 Setup

Only for 360° videos that were exported with Direction Lock (Insta360 Studio) or World Lock (GoPro Player). In such a video the picture centre stays on a fixed compass direction, so the camera no longer turns with the road – you look sideways in every bend. "Follow route direction" turns the camera into every bend again, using the course from the synchronised GPX track. With a normal 360° export the picture centre already follows the camera; leave the function off there, it would turn the view away from the road. The app asks this once when you switch it on.

The app has to learn once per video where "forward" is: put the marker anywhere in the video, drag the picture so that you look straight ahead along the road, open Config → 360 Setup and press "Forward is here". Confirm the question; the calibration can be undone with CTRL+Z. Videos that are not calibrated keep their keyframes. Smoothing (seconds of track averaged on each side, default 2 s) calms the course in bends, Look ahead (default 2 s) makes the camera look where you will be in a moment, like a rider does. With the function on, keyframes only set tilt and zoom; the direction comes from the track. Settings and calibration are stored in the project; the export needs no GPX. The dialog also has "Horizon tilt" for the current video: a correction in degrees for a 360° export whose horizon is slightly off (positive turns clockwise). The picture follows while you change the value; OK keeps it, Cancel restores the old value, CTRL+Z undoes it. It is stored in the project and applied in the export.



AutoCutVideo+GPX

To activate this option, Edit Video must be enabled first.
This is one of the most important features of the program!

- When this option is activated, selecting a section of the video for cutting (e.g., a stop at a traffic light or the start of the video) will automatically cut the GPX file as well!
- This allows you to remove breaks from both the video and GPX with minimal effort, ensuring the project remains synchronized.

Player Setup

- **Show Endcut Warning**
  shows a pop-up when the player reached a Endcut (you set a cut on the end of the video)

- **Time: Global/Final**

  The timer at the top right of the video, which shows the current video time, can be switched between Global and Final mode.

  What is the difference?

  - Global time: The total duration of all videos before editing.
  - Final time: The total duration after cuts have been applied.

  For example, if a video originally lasts 10 minutes and you cut out 2 minutes, then:

  - Global = 10 min
  - Final = 8 min

  Why is this useful?
  If you're editing a long route with multiple stops and breaks, it's best to review all videos quickly in an external player and note down break times.

Example:

- The first video has a break at 4:31
- The second at 2:20 and 4:40
- The fifth at 7:10

When you load all videos and cut the first break, it becomes difficult to calculate the exact position of the next break.
Using Global time, the second break remains at (length of video 1) + 2:20, making it easier to locate without extra calculations.
This is especially useful for GoPro files, which usually have the same length (unless manually stopped).

Player-Setup

- *Show Endcut*
  ON: Player shows the Final-End of the Video (shows last Frame of Final Video)
  OFF: Player jump to the Global-End of the Video (ignore cuts)

- *Show Endcut Warning*
  ON: You get a warning or OFF:not

FFMPEG

ffmpeg is not part of KVRouite. It is only needed for Copy Mode, which cuts on keyframes with ffmpeg and uses ffprobe to index them.

- Normally there is nothing to do here: if ffmpeg is in your PATH, KVRouite finds it. Use these entries to show which path is in use, to point KVRouite at a different folder, or to forget a stored one.
- Without ffmpeg and ffprobe, Copy Mode stays greyed out. Nothing else is affected.

Temp Directory
The Encode-Mode need a lot of HD Space and imn case you haven't enough on the actually Partition, you can choose another Directory!

Chart Settings

Limit Speed
Adjusts the maximum peak in the speed curve.
It is not uncommon for a GPX point to show an unrealistic speed (e.g., 240 km/h) due to recording errors, signal loss, or faulty cuts. If the speed limit is not set (e.g., to 70 km/h), the curve would be flattened by extreme peaks. This setting ensures that speeds above the limit (e.g., 240 km/h) are capped at 70 km/h (or your chosen value).

Mark ZeroSpeed

Marks GPX points with a downward orange line in the diagram when the speed drops below a set threshold. The default threshold is 1 km/h. If a GPX point falls below this speed, an orange marker is displayed. This helps quickly identify pauses in the GPX file.

Mark TimeGaps
Another function for identifying breaks. GPX points with a time gap greater than the specified value (e.g., >1s) will be marked in the diagram with blue

Map Setup

Directions:
If you have a Mapbox key/token set up, you can use the Directions function to draw a route that follows known roads and paths, just like a route planner. Two new buttons will be added to the map, allowing you to choose whether the route should be drawn for cars, bicycles, or pedestrians. Additionally, the "Close Gaps" function (see below) will also change accordingly.

Map Keys

You can enter API keys for different tile providers (terrain maps).
Without a valid key, terrain maps cannot be displayed.
Most providers offer a large number of free tiles per month. At the time of this software's

release, Mapbox provided 50,000 free tiles per month—a limit that most users will never exceed.

Points Size:
The size of GPX points on the map can be adjusted.
This is useful when points are closely clustered together.

Sync all with video

When activated, many actions are synchronized with video:

- New points inserted at current video time. If Directions is activated and Mapbox provided, a route is created to the new point (after ask transport mode).
- When we select a point the video jumps to the corresponding time

Lock Window Width

Buttons appear and disappear while you work – when Video Edit is switched on, for example, or when a longer value shows up in the info line. Qt then widens the window on its own and the layout jumps.

With this option on, the window is measured once in its widest possible state – all buttons visible, longest texts – and that width is kept as the minimum. Nothing jumps any more. The window can still be enlarged by hand, it just cannot be made narrower than the toolbars need.

⚠️ Important: on a small screen the reserved width can be wider than the screen itself. Leave the option off there.

The setting is remembered between sessions.

GPX Bar Position
Where the GPX button bar and the info line (video time, GPX length, elevation, slope, zero and gap counts) are shown. Five places: Under GPX Table (the default), Over GPX Table, Under Map, Over Map, and Floating. It is the same bar everywhere; at the map and floating it is drawn a little flatter so the map keeps its height.
Floating: the bar gets a grip on its left. Drag it by the grip anywhere inside the window - over the map, the table, the chart or the video. It cannot leave the window, and it remembers its position. When it covers the bottom of the GPX table, the table leaves that much room at the end so the last row can still be scrolled into view. The choice is remembered.

Reset Window Layout

KVRouite remembers the window size, the window position and the three dividers - between the two upper windows, between the two lower ones, and between the rows - when you close it, and restores them on the next start. So you only have to arrange the window once.

This entry forgets all of that and returns to the delivery state: default size (16:9, 90 % of the screen), centred on the screen, and the splitter exactly 50/50.

Reset Config
Resets all settings to factory defaults. App restart required!

Menu GPX-INFO

GPX-Summary
Shows the summary or the loaded GPX File, like length height …

Check the following values to ensure your GPX file has no unusual peaks:

Show max Slope: Displays the GPX point with the highest slope.
Show min Slope : Displays the GPX point with the lowest slope.
Show max Speed: Displays the GPX point with the highest speed.
Show min Speed: Displays the GPX point with the lowest speed.

Average Speed
Show the average Speed of a selected range

Menu Bookmarks

When syncing a track you look at the same GPX point again and again, and you had to find it again each time: before a cut it was row 1304, after the cut it is row 1296, and chT or "Cut GPX to video" change its time. A bookmark remembers the point by its coordinates and a name - nothing else. Coordinates are the one thing no edit changes, only deleting a point removes them. So a bookmark still finds its point after cuts, time edits and smoothing.
The same menu opens on right-click on the map and on right-click in the GPX table.

Add bookmark for selected row… – Shortcut: STRG+B
Bookmarks the selected GPX row. A small window asks for a name; the default is the row number ("Row 1304"). On the map and in the table the menu names the row it will bookmark: on the map it is the point you clicked last (the blue one), in the table the row under the mouse.

Bookmarks…
Lists the bookmarks of the active slot in track order, each with its current row and its coordinates. Go to jumps to the point, Rename changes the name, Delete removes the bookmark. Double-click on an entry jumps too.

Go to bookmark – Shortcut: STRG+1 … STRG+9
Below the two entries the menu lists every bookmark. Clicking one - or pressing STRG+1 to STRG+9 for the first nine - selects its row exactly like a click in the table: the row turns yellow, the map centres on the point, the chart follows, and with "Sync all with video" the video follows too. The status bar shows row and time.
The list is sorted by position in the track, not by the order in which the bookmarks were set. A bookmark set later on an earlier point comes first, and the numbers follow that order - so a number can change when a bookmark is added in front of it.

When a point was cut out
If the exact point no longer exists, the jump goes to the nearest point within 5 m and the status bar says how far it is. If there is nothing within 5 m, nothing happens and the status bar says so.

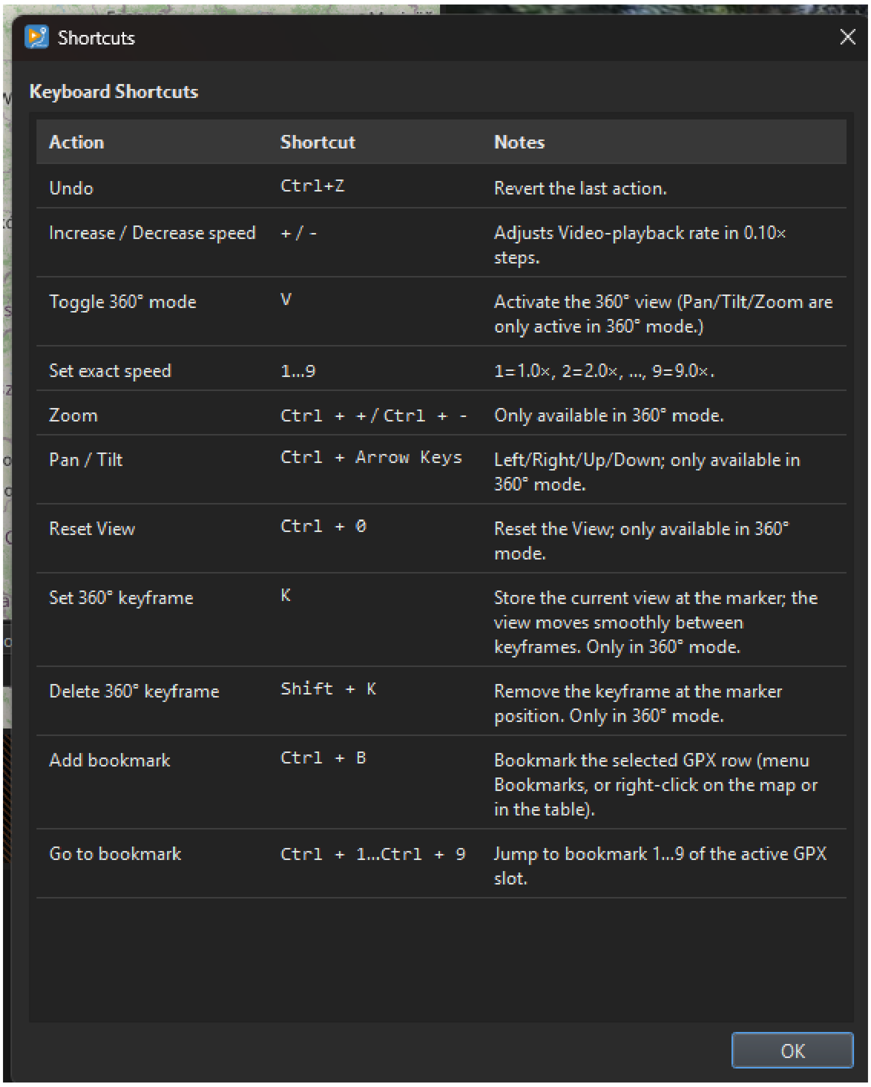
Slots and project file
Bookmarks belong to a GPX slot: Slot 2 holds a different track with different coordinates. Loading a new track into a slot clears its bookmarks. They are saved in the project file and come back when the project is loaded.

Menu Help

Show documentation

Displays this document

Shortcuts
Lists the available shortcuts:



Shortcuts

Keyboard Shortcuts

Action	Shortcut	Notes
Undo	Ctrl+Z	Revert the last action.
Increase / Decrease speed	+ / -	Adjusts Video-playback rate in 0.10× steps.
Toggle 360° mode	V	Activate the 360° view (Pan/Tilt/Zoom are only active in 360° mode.)
Set exact speed	1…9	1=1.0×, 2=2.0×, …, 9=9.0×.
Zoom	Ctrl + + / Ctrl + -	Only available in 360° mode.
Pan / Tilt	Ctrl + Arrow Keys	Left/Right/Up/Down; only available in 360° mode.
Reset View	Ctrl + 0	Reset the View; only available in 360° mode.
Set 360° keyframe	K	Store the current view at the marker; the view moves smoothly between keyframes. Only in 360° mode.
Delete 360° keyframe	Shift + K	Remove the keyframe at the marker position. Only in 360° mode.
Add bookmark	Ctrl + B	Bookmark the selected GPX row (menu Bookmarks, or right-click on the map or in the table).
Go to bookmark	Ctrl + 1…Ctrl + 9	Jump to bookmark 1…9 of the active GPX slot.

OK

STRG+B adds a bookmark for the selected GPX row, STRG+1 to STRG+9 jump to bookmarks 1 to 9 (see Menu Bookmarks). K stores a 360° keyframe at the marker, SHIFT+K removes it (same as the KF button and Del on a selected diamond).

Youtube tutorials

Opens Youtube channel in browser

Privacy

Shows which connections KVRouite makes over the network, and when. The same text is shown once when the program is started for the first time.

Check for Updates

Asks GitHub whether a newer release exists. Only the program version is sent.

Auto Check for Updates

When this is ticked, that check runs a few seconds after every start. Switch it off and KVRouite contacts GitHub only when you ask it to.

Copyright

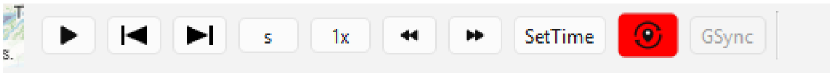
Displays copyright information and the installed version, lists the third-party components and names the folders that hold their full license texts: _internal/gstreamer, _internal/qt and _internal/third-party-licenses.

Video Control Buttons

Note: Depending on the mode, not all buttons are visible.
Some buttons only appear after Video Edit is enabled, while others change functionality when AutoCutVideo+GPX is activated.

Here are the two different views:

Edit Off:



Edit On:

Video Control Buttons

The buttons from left to right:
Play/Pause – Go to start – Go to end – Step-Value – Multiplier Stepper – Step Left – Step Right – SetTime – MarkB – MarkE – Deselect [x] – Cut – KF (360° only) – CutBegin – CutEnd – Define GPX/Video sync point – GSync – Overlay [Ovl] – AutoCutVideo+GPX [VG]

Play/Pause
Starts and pauses the video.

Go to start
Resets video position to the beginning.
*hidden feature: Right-click jumps back to the previous cut edge or video merge; at the start it wraps to the end

Go to end
Sets video position to the last frame.
**hidden feature: Right-click to navigate through all cuts and video merges*

Stepper
The stepper consists of the following buttons: Step-Value – Multiplier – Step Left – Step Right

Stepper
With Step-Value, you can select the type of step: seconds (s), minutes (m), keyframes (k), frames (f), cuts (c), or – in 360° mode – the keyframes of the viewing direction (KF). Important: Keyframes can only be used if you selected "Yes" when asked about Indexing Keyframes.

Cuts (c)
The step value c walks from one cut edge to the next. Every cut gives two stops: the last frame that stays in front of it, and the first frame that stays behind it. Between those two there is nothing left in the finished video, so they are exactly the pair that will sit next to each other. That lets you judge how a cut will look before you render anything. The first and the last frame of the finished video are stops as well. c works in Encode Mode, where a keyframe is forced on every cut so the cut lands exactly on your mark. Copy Mode cuts at the nearest keyframe instead, so the marked edges would not be what you get – pressing the arrows there tells you so and nothing moves. Use keyframes (k) in Copy Mode to see where the cut will really land.

The Multiplier determines the size of each step, e.g., 1 second (s 1x) or 8 minutes (m 8x). It has no effect on frames (f) and cuts (c) – those always move one step at a time. KF jumps from one 360° keyframe to the next; the Multiplier applies.

Left (←) and Right (→) indicate the direction in which the step should move (forward or backward).

Steps are counted in the finished video, not in the raw footage. A cut does not exist in the result, so the stepper flies over it: standing on the last frame in front of a cut, one step of 1 second takes you to one second after the cut, and one step of 1 frame (f) takes you to the first frame behind it. Keyframes that fall inside a cut are skipped for the same reason.

SetTime
Manually enter the exact time the player should display.

MarkB [-

Only available when Video Edit is ON!
Marks the beginning of a cut point in the video.

If AutoCutVideo+GPX is also activated:
Marks the beginning of a cut point in both the video and the GPX file.

MarkE -]

Only available when Video Edit is ON!
Marks the end of a cut point in the video.

If AutoCutVideo+GPX is also activated:
Marks the end of a cut point in both the video and the GPX file.

Deselect [x]
Clears the selection for MarkB and MarkE.

Cut

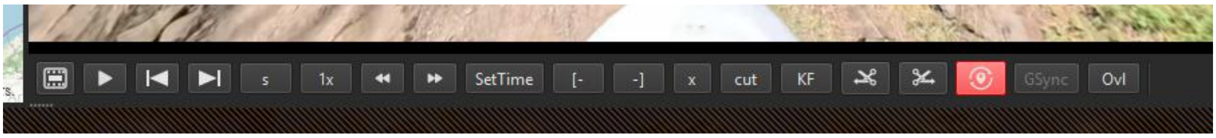
Deletes the marked section.
If Video Edit is ON: Only the selected section in the video is removed.
If AutoCutVideo+GPX is ON: The selected section in both the video and the GPX file is removed.

Note:
In the video, the removed section is black-marked and will be skipped during playback.
In the GPX file, the section is completely removed and no longer visible.

KF

Only shown in 360° mode.
Stores the current viewing direction and zoom at the marker as a keyframe, or updates the keyframe that is already there. Set the keyframe first, then drag the picture to the view you want – the keyframe takes it. Shortcut: K. SHIFT+K removes the keyframe at the marker. See Menu View → 360° Video and chapter Timeline.

The bar in 360° mode, with KF between Cut and CutBegin:



CutBegin

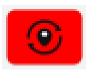
Only available when Video Edit is ON!

We synchronize the start of the video with the GPX file.

Both the video and the GPX file will be trimmed at the exact point. In the example, the first 10 seconds of the video would also be cut, and the GPX file would now start exactly at the marked point.

CutEnd
Only available when Video Edit is ON!

The GPX file is also trimmed at the same point.

Define GPX/Video sync point 

Allows to calculate the shift between video and gpx start. Find the matching gpx point corresponding to a video frame and click this button. Once set, the video and gpx edit (if video in edit mode) and selections are synchronized. New created points are inserted at time corresponding to current video time.

You can click this button again to edit the shift between gpx and video.

GSync

Active only when gpx/video sync point is defined.
Displays the corresponding GPX point for the current video position. This is useful for checking synchronization after making a cut. Set the video to a distinctive point (e.g., after a cut) and press GSync. The matching GPX point will be highlighted, allowing you to compare them. Recommendation: Check multiple points throughout the video to ensure that no pauses were missed during synchronization.

Ovl:
Add an Overlay on the marked Timeline-Position. A Window will open and ask which Overlay you want to add from the pre-configured one (you can add a new one too).
Set the Duration (max 30s) and the fade-in and fade-out time.
Please use jpg-files only.

VG✓/x
Same as Confif Menu AutoCutVideo&GPX On and Off

Map Control Buttons

New point now can be in 2 modes:

: icon when “Sync all with video” is off
Adds a new GPX point on the map, either at the beginning or end of a route. Alternatively, you can place it precisely on the line between two existing points.

- The route will automatically be extended by 1 second.
- Before adding the new point, you must select the point where it should be connected.

: icon when “Sync all with video” activated

Adds a new GPX point on the map synchronized with the video

- With timestamp corresponding to the current video position.
- Inserted in the list in the according position, based on its time

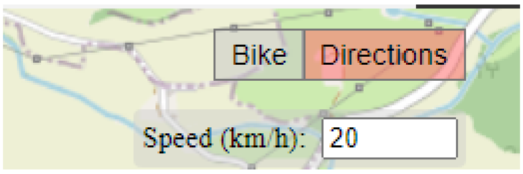
Move
Activates the "Move GPX Point" mode. This allows you to freely move a GPX point on the map. Useful for adjusting the route to align with roads if the recorded path is incorrect.

V-Sync
Displays the video frame corresponding to the selected GPX point. For example, if the GPX position is at 10 seconds, pressing V-Sync will show the frame at 10 seconds in the video. It is recommended to press the button multiple times to ensure the correct frame is displayed, as system performance may cause delays. Keep in mind that not every exact frame can be displayed—only the nearest displayable frame. There might be a slight millisecond discrepancy, as some frames are not directly accessible due to GOP (Group of Pictures) encoding.

In "Sync all with video" mode this button is visually activated and triggered at any point selection.

View-Switch
Toggles between different map display modes: OSM (OpenStreetMap) - Various satellite views
Note: If no API key is entered for services like MapBox, only OSM will be displayed. To access terrain maps, you must request an API key from the respective service provider.

Directions-Control-Buttons:



For Experts ONLY!
The "Bike" button toggles between Bike, Car, and Foot. Depending on the selected mode, the route will be requested and calculated accordingly via Mapbox.

Function:
First, you need to select either the first or the last point of the route by clicking on it. Then, activate the Directions function (its color will change). Using "Speed," you can define the spacing between points, and with the crosshair, you can set the target point where the route should lead. This can only be done at the beginning or end of a route!

It is crucial not to select excessively large "steps" but to create the new route in small increments instead. Additionally, always verify the connection to the existing route, as the "old" point often needs to.

If you want to recalculate a section of an existing route—for example, if your GPS device did not record anything due to a lost signal or any other reason—you can use the **Close Gaps** function while Directions is active.

The functionality is the same as **Close Gaps**, but instead of drawing a direct path, the route will be adjusted to follow the road.

Right-click on the map
Opens the Bookmarks menu for the point you clicked last on the map (the blue one): "Add bookmark for row N...", the list of bookmarks, and "Bookmarks...". See Menu Bookmarks. The browser menu (Back, Reload, Save page...) that used to appear here is gone.

GPX Control Buttons

The buttons from left to right:
markB – markE – Deselect [x] – Delete – chTime – chElevation – chSlope – More (...) – Smooth

markB
Marks the beginning of a cut in the GPX list. The marked section will be highlighted in red in both the GPX list and the map. The button turns red when activated.To delete a single point, simply mark it with markB.

markE
Marks the end of a cut in the GPX list. The entire section between markB and markE will be highlighted in red in both the GPX list and the map.

markB / markE while the video runs
With AutoCutVideo+GPX off, the two buttons take the selected row. If the video has moved on - the yellow row is not the selected one - the app asks whether to set the mark on the row under the video. Yes selects that row (blue on the map, table and chart follow) and sets the mark; No leaves the marks unchanged. With nothing selected and no yellow row, the status bar asks you to select a row or a point on the map first.

Right-click on markB / markE
After a cut - Cut or Remove in the GPX bar, or the video cut with V&G on - the marks are gone, and the next step is usually on the gap the cut left. Right-click on markB marks the point before the removed range, right-click on markE the point after it, and shows it: row selected, map centred on it, chart on it, and with "Sync all with video" the video too. Only in the GPX bar with V&G off. The gap is remembered as long as the track keeps its point count; after another cut, Undo or a new track it is forgotten.

Right-click in the GPX table
Opens the Bookmarks menu for the row under the mouse: "Add bookmark for row N…", the list of bookmarks, and "Bookmarks…". See Menu Bookmarks.

Deselect [x]
Deselects a marked section or a single selection.

Cut (scissors)
Deletes an entire marked section. To prevent a gap, the time between the two surrounding points is adjusted to 1 second. Points next to deleted section will have their time shifted back by the amount corresponding to the deleted section length.

Remove (minus button)

Also deletes the marked section, but the next points time will not be shifted. We can see this as a route simplification, with less points. (needed for rebuilding a route with Close Gaps or by hand)

chT (chTime)
Modifies the time or step of a single point or all points in a marked section. Useful for fine-tuning synchronization. This button allows to edit the step of the selected points in the list. Note that next points will be shifted by the time difference. For example if you set 2s instead of 1s on a single point all the next points will be shifted to +1s. If you don’t want the following points to change edit the time value directly in the list (by double click).

chEle (chElevation) Changes the elevation of a single point or an entire marked section. This is useful when there is a sudden elevation shift due to a GPS restart after a break. If the GPS records a different elevation after restarting, the entire section can be shifted so that: The first point after the restart has the same elevation as the last point before the restart.

%↗ (chSlope)
Adjusts the slope (gradient) of a single point or a marked section. Used to correct elevation recording errors or smooth transitions after a cut.

Undo
Reverts the last action.

Smooth
Opens a smoothing settings window, where adjustments can be made to smooth the elevation profile. Default settings usually provide a good result.If the result is not satisfactory, the options can be adjusted: Box-Smoothing: Number of GPX points included in the smoothing process. Flatten-Value: Maximum slope difference between points.

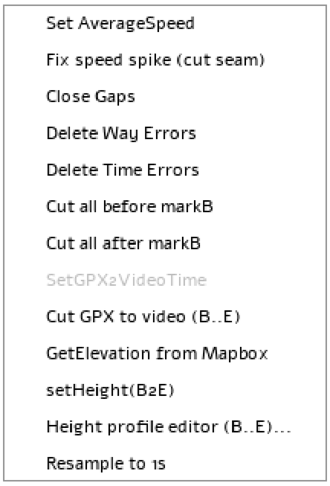
Slot 1/2
switch between Slot 1 and Slot 2
Read more in the Section Slots

Slot Sync

Smooth: range, warning, climb
With markB and markE set, only that range is smoothed; the neighbours outside go into the averaging so the edges do not kink, and the heights of B and E stay. Without marks the whole track is smoothed - the dialog says so in its first line, with the point count - and the end point stays too. The program remembers the last smooth: run it again on the same heights and it warns, with time, parameters and the climb before and after; change some heights in between (Height profile editor, chEle) and it finds those rows itself and offers to smooth only them. The result message shows the climb of the track before and after.

See chapter Slots

More-Menü […]:



Set Average Speed
Calculates the average speed of a selected section. This can be helpful if a speed peak cannot be corrected in another way.

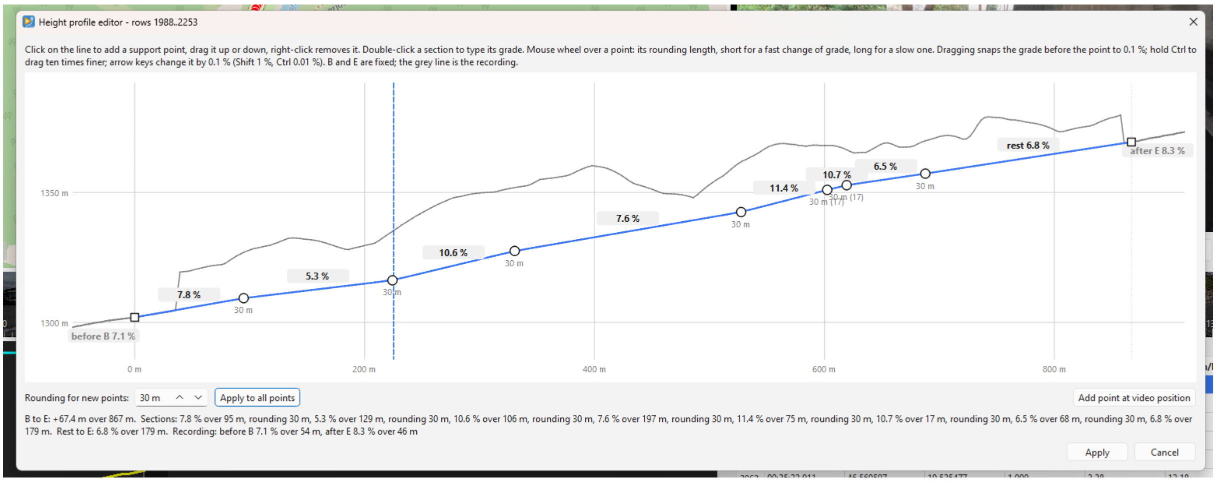
Fix speed spike (cut seam)
When a camera failure has been cut out of the video, the cut also removes the GPX points of that time and shifts the following points forward. The distance ridden in the meantime remains as one single segment, for example 105 m in one second = 380 km/h. Kinomap and other platforms calculate the speed from the GPX file themselves, so the spike would show up there as well. Set Average Speed is not the right tool for this: a constant speed across changing gradients does not look like a real ride.

The function works on the selected row. If that row is not a spike, it lists all spikes found in the track (a segment longer than 20 m that is at least three times faster than its surroundings) and lets you pick one. The dialog shows the diagnosis: distance, time and speed of the segment, the median speed of the 60 points on each side, and the riding time that is missing. The proposal gives the seam this missing time and takes the same time back from a range of points before and after by dividing every step there by one factor. All speeds in the range rise by this factor in their real proportions, the flatter part stays the faster one. Positions, elevations and gradients are not changed, and no point after the range moves, so the video stays in sync.

The range is chosen automatically: 60 points per side, growing in steps of 30 while the factor is above 15 % and the gradient of the added points stays within 2 percentage points of the gradient at the seam. It stops at the ends of the track and before another spike; the two sides may differ. Both sides can be changed in the dialog, and the table is recalculated immediately: gradient, speed before and after for the ten points nearest the seam on each side, then per 60 points, the seam itself and the maximum in the range. A chart below the table draws the speed of every point in the range - grey before, coloured after - with the seam as a dashed line, so you can see that the seam is not slower than its neighbours; the spike itself is clipped at the top. Apply takes a GPX undo snapshot. As with chT, video cuts can no longer be undone afterwards, so set all cuts first and fix the seams last.

Cut GPX to video (B..E)
For a merge edge where the camera had been off: the video cut takes only the seconds around the edge out of the track, the time the camera was off is still in it, and from the seam on the track runs ahead of the video. V&G off. Set markB on the seam (right-click markB after the cut), drive the video to a landmark you can identify, mark its GPX point as markE, then choose the entry. It takes the table time of markE minus the current video time - the time the camera was off - and removes exactly that much track after markB; everything behind moves up, and markE lands on the video time. The seam keeps the distance ridden meanwhile, so right-click markB and markE and run Fix speed spike afterwards. Refused when the point is not later than the video or the cut would reach markE.

Height profile editor (B..E)...
In a half-open gallery or on a viaduct even a barometric bike computer records nonsense, and no terrain model helps there - it measures the roof or the slope under the bridge. What is known: the heights at both ends, the distance, and what you see in the video. Set markB and markE on two trustworthy points and open the editor: a window with the profile of the section, distance against height, the recording grey in the background, B and E fixed, and a stretch of the recording before B and after E with its grade.



Click on the line to add a support point, drag it up or down, right-click removes it. Dragging snaps the grade before the point to 0.1 %; hold Ctrl to drag ten times finer; the arrow keys change it by 0.1 % (Shift 1 %, Ctrl 0.01 %). Double-click a section to type its grade. Every section shows its grade, the last one what is left to E - so the first point already shows whether the rest can work out.

Rounding: the kink at each support point is replaced by a vertical curve. Its length is set per point - mouse wheel over the point, or the field below with the point selected - long where the grade changes slowly, short where it changes fast. "Apply to all points" sets one length everywhere. The length is limited to the shorter neighbouring section, so two roundings never overlap; the value under the point shows the effective length in brackets when it is limited.

The window does not block: the video position is a dashed line in the profile, "Add point at video position" adds a support point there, and a click on a support point drives the video to it. Apply writes the heights of B..E only, as one undo step; Cancel leaves everything as it was. Typical workflow: watch the video, stop where the grade changes, add a point there, drag it until the grade reads what you see, check the rest to E, then Apply. Smooth afterwards on the same range if needed - Smooth offers exactly those rows.

Close Gaps / Close Gaps (Directions)
Fills in missing GPX points. Example: You drive through a tunnel or an area without GPS coverage.The GPX file shows an unusually large time step, and the chart displays it as a pause (since the step is greater than 1s). Cause: GPS signal failure. **Solution:** Using Close Gaps, the missing section is filled with GPX points at 1-second intervals. No changes are made to time or distance—it simply looks better in the visualization. If necessary, the new points can be adjusted to match the road.

Important: markB must be set before the gap. markE must be set after the gap.

Close Gaps (Directions):
When Directions is activated, GPX points are no longer simply connected in a straight line from markB to markE. Instead, you can let Mapbox draw the route along the road.

The transport-mode dialog (Bike, Car, Foot) has the checkbox "Get elevation from Mapbox", checked by default and remembered. Checked, every new point on the road gets its height from the Mapbox terrain tiles. Unchecked, no terrain is fetched: the height runs in a straight line from markB to markE at one constant gradient - like Close Gaps without Directions, only along the road. Inserting a point on the map with Directions on uses the last choice made in the dialog.

Delete WayErrors
When loading a GPX file, you will receive a warning if WayErrors are found. These errors are also highlighted in red in the chart. This button fixes these errors automatically. What are WayErrors? These are GPX points that have identical latitude/longitude but a 1s time step—a recording error. Fixing the error recalculates the second point: It is moved to the middle between the previous and next point while keeping the same latitude/longitude.

This should always be executed!

Delete Time Errors
This issue is rare, but some GPX points might have a 0s time step.These points are simply deleted.

This should always be executed!

Cut all before markB
Deletes all GPX points before and including the markB point.

Cut all after markB
Deletes all GPX points after and including the markB point.

SetGPX2VideoTime
For Experts Only!

Synchronizes the GPX time (range) with the video time (range).
This function aligns a marked time range in the GPX list with the corresponding marked time in the video editor.

Requirements:

- Video Edit = ON
- AutoCutVideo & GPX = OFF

Instructions:

1. Set the start point markB in Video_Control using markB.
2. Use GSync to select the corresponding GPX point with markB in GPX_Control.
3. Navigate the video to a second clearly identifiable point and mark it with markE.
   - This range will now be highlighted in yellow.
4. Find the corresponding point on the map and mark it with markE.
   - Now, both the video and the GPX list have a marked range.
5. Click SetGPX2VideoTime to adjust the GPX time range to match the video time range.
   - Every point in the GPX section will be recalculated.
   - The marked section in the video will now have the same duration as the marked section in the GPX list.

GetElevation from Mapbox
Retrieves elevation data from Mapbox for the selected area and stores it.
Note: You need a Mapbox Key/Token

Since 6.11 the height is interpolated between the four neighbouring pixels of the zoom-15 terrain tiles, so the values no longer come in steps with slope spikes between them. The resolution of the source itself is unchanged; in a gallery or on a viaduct it measures the terrain, not the road - use the Height profile editor there.

Set Height(B2E)
Change the final height (markE) in a selected area.
We select an area, markB and markE, and now we can change the final height, markE.
All points are increased evenly, and the subsequent points are adjusted accordingly.

Resample to 1s

In case your Steps between the GPX-Points are not in 1s Steps you can resample it.
Please notice that this change the distance / height and maybe a sync.
Please check after resample (it is recommend to this first and that's why we check it at every load of a gpx file)

If a range is marked, we only resample the marked range, otherwise the complete track!

The Info Line

Directly under the GPX control buttons there is a line with short values. The labels are kept short on purpose so the window can still be split 50/50 on a small screen. Hover over any of them and a tooltip explains what it is.

V: 00:04:25.565

The total length of all loaded videos, with your cuts already deducted. This is what the finished video will be, not the current playback position.

GPX: 8.94km/00:04:25.565

Length and duration of the GPX track – first the distance in kilometres, then the time from the first to the last point. Compare it with V: to see whether video and track still match after cutting.

Ele: 244m

Elevation gain: the sum of all climbs in the track. Descents are not subtracted, so this is the total metres uphill, not the difference between start and finish.

Max%: 4.7%   Min%: -27.1%

The steepest climb and the steepest descent in the track. Extreme values here are a good hint that the GPX has a bad point – a slope of 40 % on a road is usually a GPS error, not a hill. Use GPX-Info > Show max Slope to jump straight to that point.

Zero: 0

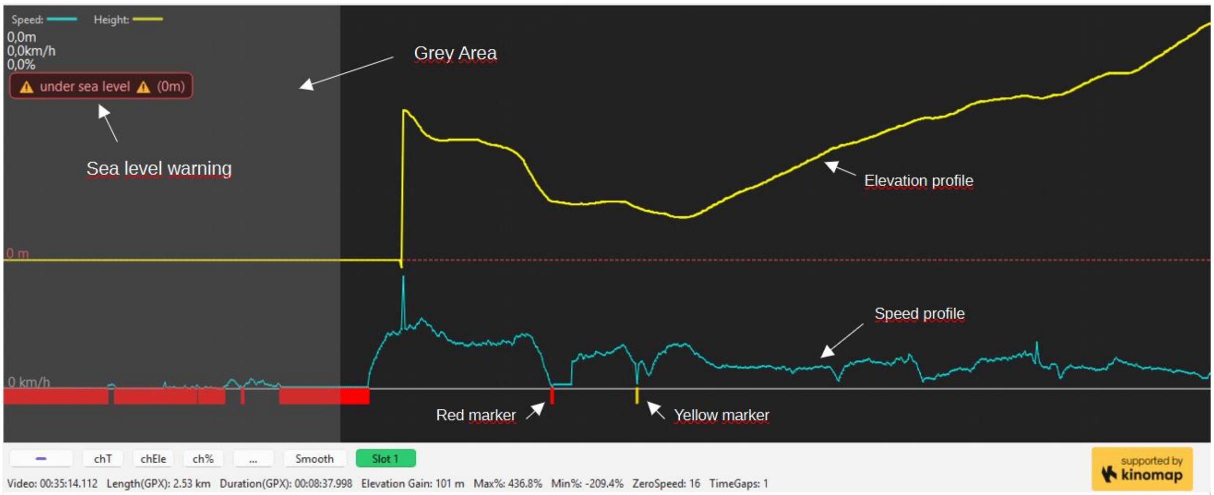
The number of GPX points whose speed is below the zero-speed threshold – in other words, where you were standing still. The threshold itself is set under Config > Chart-Settings > Mark ZeroSpeed. These points should be cut out: a Kinomap video should not contain a stop at a traffic light.

Gaps: 0

Time gaps: the number of places where two consecutive GPX points are more than one second apart. That happens when the GPS lost its signal, for example in a tunnel, or when the recording was paused. A gap is not automatically wrong – if the video shows you moving and the point has a sensible speed, everything is fine. Config > Chart-Settings > Mark TimeGaps marks them in the diagram, and Close Gaps in the More menu […] can repair them.

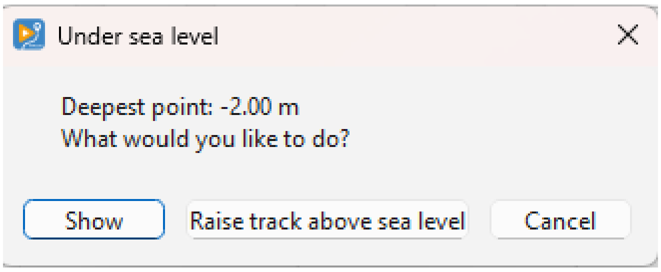
As long as Zero and Gaps are both 0, the track is clean.

The Chart:



you will see:

- **The elevation profile at the top**
  show the elevation Profile of the Route
- **The speed profile at the bottom.**
  show the speed profile
- **Sea level warning**
  if there is a point under sea level (under 0m) we show this warning,
  you shouldn't upload a route with a sea level warning
  Right Click on the warning: we show step by step every point under sea level
  Left Click on the Warning:



You get a information which is the deepest point and you can choose between:
- Show (will show the point)
- Raise track above sea level – we move all points over the sea level with exact the shown value and the warning should be gone, the profile will be the same but now over 0m

- **Red Marker**
  this will show the ZeroSpeed Points and it’s a clear marker that you are stopping here and have 0km/h speed (you can setup the value in the Setup Menu). This Points must be cutted! We don’t want ZeroSpeed Points. If you take a look at the info line directly under the buttons you can see how many of them you have: Zero: for ZeroSpeed points, Gaps: for TimeGaps (see chapter The Info Line)
- **Yelllow Marker**
  this shown points with higher Steps than 1s (you can change the value in the Setup Menu) Check this points if there is something wrong, but sometimes there isn’t nothing really wrong, because you was riding through a tunnel and lost the signal, than the first point after the tunnel have a TimeGaps. If yu play the Video at this point and your are moving and the point has a Speed value all is ok. Of course we can fix that with our tool (Close Gaps and some points more, but for that: follow the Tutorials.

- **Grey Area**
  If you have uses the sync button and the first part of the gpx-list, this area is grayed too and this mean it will not be written to the exported gpx file, all red and yellow markers are not shown in the info-list and you can ignore them. You see o sea level warning if a point is in the grey area and under sea level!

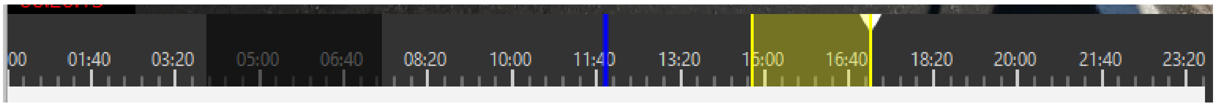
Navigation Controls:

- Zoom: Use CTRL + Mouse Wheel.
- Pan: Hold the right mouse button and drag to move the chart.

Zoomed in

At zoom 1 the height axis spans the whole track. As soon as you zoom in, both axes follow the points in view: the height axis covers the visible minimum and maximum with a margin (never less than 2 m of span), the speed axis keeps 0 at the bottom and follows the highest visible speed at the top. The upper and lower values of the window are written at the right edge, so you can tell whether a step is 20 cm or 20 m. Once the points are 4 pixels or more apart they are drawn as small circles, so every recorded point is visible. The zoom limit follows the number of points: at full zoom one point takes 40 pixels, about 25 points across the chart.

Timeline:



Visual Representation of Video Time

- Blue Marker: Indicates the separation between two videos. Right-click it for a Merge-Fade (see below). A join with a Merge-Fade is drawn green, line and fade wings alike.
- Black Area: Represents an already deleted section (in Encode Mode, with right click into the black Area you can switch this cut between crossfade and hard cut)
- Blue Area: Marks a Overlay (with right click into the blue Area you can delete the overlay)
- Yellow Area: Marks a selected section.

Hard Cut instead of a crossfade

In Encode Mode every cut is bridged with a crossfade, using the XFade duration from the Encoder Setup. For one particular cut that may not be wanted – for example when the scene changes completely and a soft blend looks wrong.

Right-click the black block of that cut and confirm the question. A cut set to hard is drawn with orange edges, so you can see at a glance which cuts will blend and which will jump.

The cut position and the total length of the video stay the same. Only the transition changes: instead of blending the material before and after the cut, the video jumps directly.

Copy Mode has no crossfades at all, so there is nothing to switch there. The first and the last cut cannot be switched either – they are trimmed away before encoding and never had a crossfade.

The setting is stored in the project file, so it survives saving and loading, and it can be reverted with Undo (Ctrl+Z).

While rendering, the encoder window lists every cut, so you can check what will be produced:

[INFO] Cut 12.00s - 18.00s: hard cut (no crossfade)

[INFO] Cut 27.00s - 30.00s: crossfade 2.0s

Merge-Fade at a video join

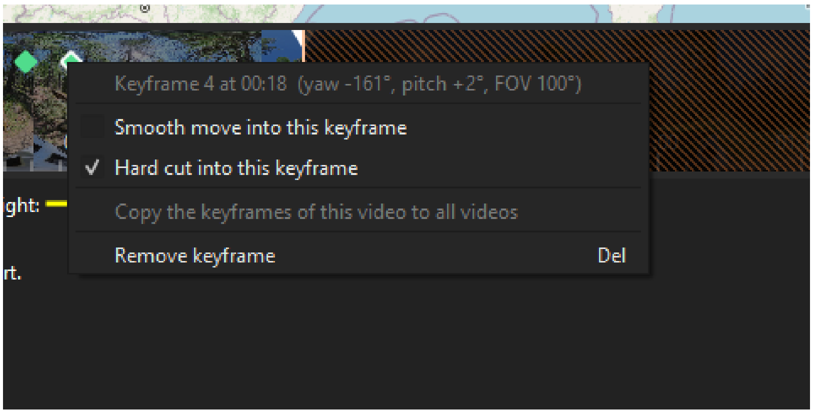
Two merged videos meet at a join without any overlap: the first one has ended, the second one is only starting. A crossfade needs material on both sides of the edge, and at a join there is none - blending would either cost seconds of ride or blend a frame into itself. The Merge-Fade bridges the join without losing anything: in the last half of its length before the join the first frame of the next video fades in as a still, from 0 to 50 %; after the join the last frame of the previous video fades out from 50 to 0 %. On the join both sides show the same mixed picture, so there is no jump, no black and no slow motion. Nothing is cut, the GPX track is untouched, and the output is exactly as long as the sum of both videos.

Right-click the blue join line in the timeline. The hit zone is 10 pixels either side of the line at any zoom; the line lights up under the pointer and the cursor becomes a hand. The menu offers "Merge-Fade", "Hard join" and "Merge-Fade length...". The length defaults to the X-Fade from the Encoder Setup and is capped at what fits up to the next cut or the ends of the video. A join with a Merge-Fade is drawn green, with the fade wings on both sides of the line; a hard join stays blue.

Whether a join gets nothing, a Merge-Fade or a cut is your choice: nothing gives a hard join, a Merge-Fade gives the transition above, and a cut across the join behaves like any other cut, with or without crossfade. Encode Mode only, like every crossfade; Copy Mode passes the material through unchanged. The setting is stored in the project file and can be reverted with Undo (Ctrl+Z). Before the preview can show a Merge-Fade, the two still frames are taken from the videos; the preparation window says which one it is working on.

360° keyframes

In 360° mode every keyframe of the viewing direction is drawn as a small green diamond at the top of the timeline. Click a diamond to select it and jump there; hold Ctrl and drag it to move the keyframe (the picture follows while you drag; a plain drag always moves the white marker, so the two never compete); Del removes the selected keyframe, Esc clears the selection. Right-click a diamond to decide how the keyframe is reached: "Smooth move into this keyframe" (the camera pans from the previous view, hollow diamond) or "Hard cut into this keyframe" (the previous view holds and jumps here, filled diamond), or remove it. All of it can be undone with CTRL+Z. Keyframes belong to the video they lie in: when videos are removed or reordered they move with their video. The right-click menu also offers "Copy the keyframes of this video to all videos": the keyframes are transferred relative to each video's start and replace the keyframes there (with confirmation, undo possible).

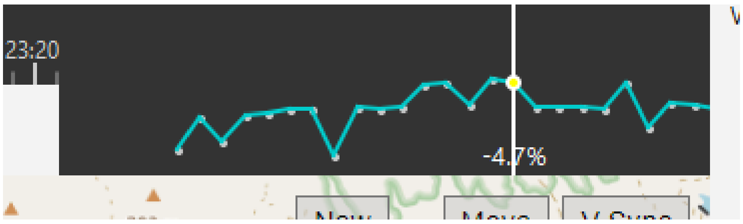


Controls:

- Zooming: Use CTRL + Mouse Wheel.

- Panning: Hold the right mouse button and drag to move the timeline.
- Clicking on the timeline: Moves the white marker (which indicates the current time) to that position and updates the video accordingly.
- Scrubbing: The white marker can also be dragged to scrub through the video.

Chart Flow:



The Chart Flow is one of the four modules and is used to check the gradient at locations such as hilltops. Unlike the big chart it scales its elevation axis only over the points currently in view, so a crest fills the frame instead of being a flat line. The marker stays in place and the points travel underneath it. Ctrl and the mouse wheel set how many GPX points are shown.

After smoothing, this chart provides a clear visualization of whether the video aligns correctly with the gradient changes.

This helps ensure that elevation changes in the GPX file match what is visible in the video.

Slots

We have now two different Slots: Slot 1 and Slot 2
if you load a GPX od FIT the files will automatically loaded in Slot 1
if you extract a GPX from a Gopro it will be loaded in Slot 2
you can work in each Slot like you want it make no changes in the other Slot.
If you prefer to work with GPX from a Wahoo/Garmin/Strava/whatever (what I recommended) you can worl like before.
but if you only have a Gopro with GPS, you had in the back to extract the gpx with a extern tool to load it into the App, now ypu can do it directly in the App

But why Slots?
for a maximum easy syncing.
Gopro-GPX are always perfect synced to the Video!
How ?
If you have a Gopro File with GPX and want sync your extern GPX loaded in Slot 1:
1. Load the GPX

2. Load the Video / Videos
3. Extract the first Video
4. You are in Slot 2 now
5. Search a point you can simple identify in the Video and in the GPX (corner/ roundabout…)
6. The magic starts now with: click on the Slot Sync - Button
7. The app search the nearest GPX Point in Slot 1 (Lat/Lon) and will show you this point
8. The Video is still on the same point
9. If the point is the same (and it is)-> press the Sync-Button
10. Synced !

Background- Information to point 9:
You have two different GPX records, sometimes the GPX we show in Slot 1 match perfectly.
Sometimes there are some meters difference, than you can simple move the Video a lttle bit forward or backward to the point we shown (only some 1x frames cklick) and it will match

Sound

Up to 6.14 the Encode-Mode wrote videos without a sound track. Since 7.0 the track goes into the export, with the same crossfades as the picture, and two tools work on it: a damper for passing vehicles and a voice remover. Everything here is in Encode-Mode; the damper and the audio view are pure GStreamer and work in every build, the voice remover and its detection need the Voice add-on.

The preview still plays the original sound - no damper, no voice removal. Only the export carries the treated track; to check a single spot before exporting, use Listen on a band (below).

Lite and the Voice add-on

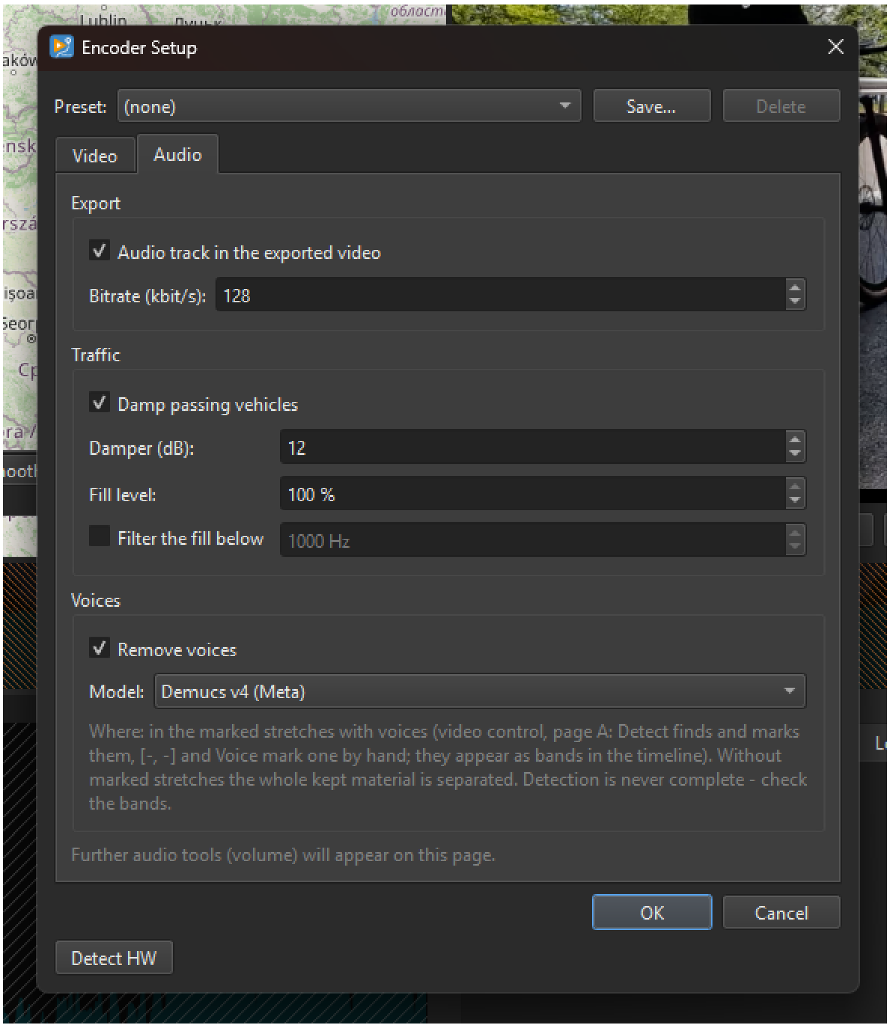
The Windows download comes in two parts. **KVRouite_<version>_Win_x64** is the application - sound track, vehicle damper and the audio view of the timeline. **KVRouite_<version>_Voice_Win_x64** is the voice remover and its detection, about 900 MB unpacked (PyTorch and the models), which most users do not need.

Install the add-on in one of two ways. The Voice **zip** carries the same folder name as the application and is simply unpacked over the existing KVRouite folder. The Voice **installer** does the same into an installed copy of the **same version** and refuses any other. Nothing else has to be set up.

Without the add-on, **Remove voices** is locked and a note says which file to fetch; the sound track and the vehicle damper still work. **--selftest** on the command line reports which of the two is installed. The macOS bundle is the application only.

The Audio page in the Encoder Setup

Encoder Setup (Config menu, Encode-Mode only) has a second tab, **Audio**:



Audio track in the exported video turns the sound on, **Bitrate** sets the AAC quality (32-320 kbit/s, default 128).

Damp passing vehicles switches the traffic damper on. **Damper (dB)** is how far a marked stretch is turned down (3-30, default 12); **Fill level** is how loud the ride noise laid over it is (0-100 %, default 100 - at 100 the spot stays about as loud and only the vehicle is gone, at 0 it just gets quieter). **Filter the fill below** (off by default) keeps only the low part of the copied piece, in case a mount squeaks.

Remove voices switches the voice remover on and **Model** chooses it: MDX-Net (the Ultimate Vocal Remover model, default) or Demucs v4 by Meta, which leaves a little more of the ride noise. Needs the Voice add-on.

The same three switches - Audio, Damp passing vehicles, Remove voices - are also on the export dialog, to flip right before encoding (below).

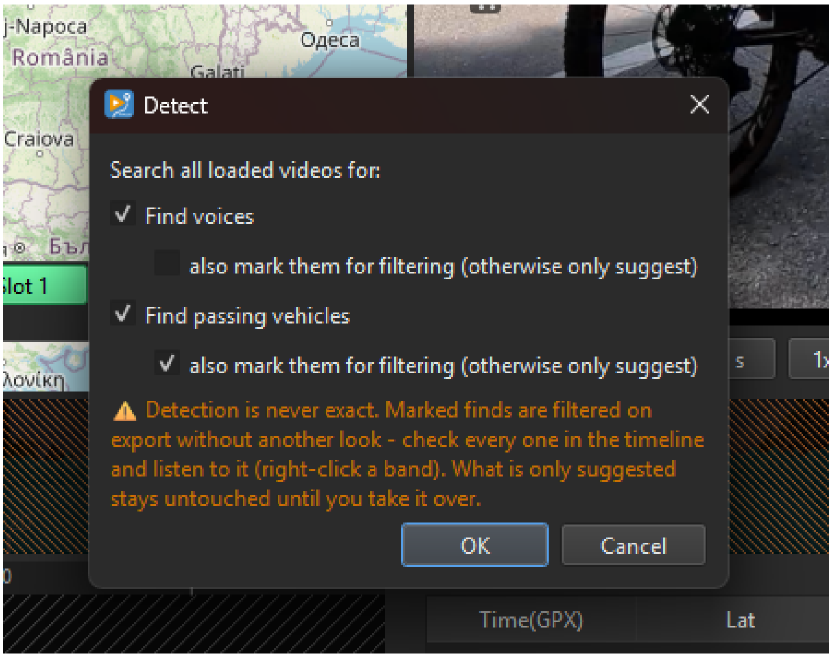
Marking where to treat - page A of the video control

Which stretches are treated is decided by ear, not by a scanner. In Encode-Mode with a sound tool on, the video control gets a button at its far left, a film strip that turns into a speaker; it flips the button row between **page V** (the video controls as before) and **page A** (the sound). Page A shows [-, -], x and, depending on what is on in the Encoder Setup, Voice, Vehicle, Detect and Sens:

Voice marks the area between [- and -] as a stretch with voices (orange), **Vehicle** marks it for the traffic damper (blue). **Sens** sets the sensitivity of the voice detection, 1-7 (1 clear speech only, 7 everything that might be, default 3). With only one tool on, only its button and Detect are shown.

Detect - find voices and vehicles

Detect searches all loaded videos at once and opens a dialog:



Tick **Find voices**, **Find passing vehicles** or both (whichever the Encoder Setup allows). Under each is a switch **also mark them for filtering (otherwise only suggest)**. Left off, the finds appear as **suggestions** - dashed frames, orange for voices, blue for vehicles - that do nothing on export until you take them over. Switched on, the finds become solid bands that are filtered right away, and a warning reminds you that a marked find is treated without another look.

Voices come from a trained detector (Silero VAD), vehicles from the sound level. Detection is never exact - a conversation in the wind can be missed, a loud pass taken for speech - so every find is meant to be checked in the timeline or the Audio Zoom and listened to.

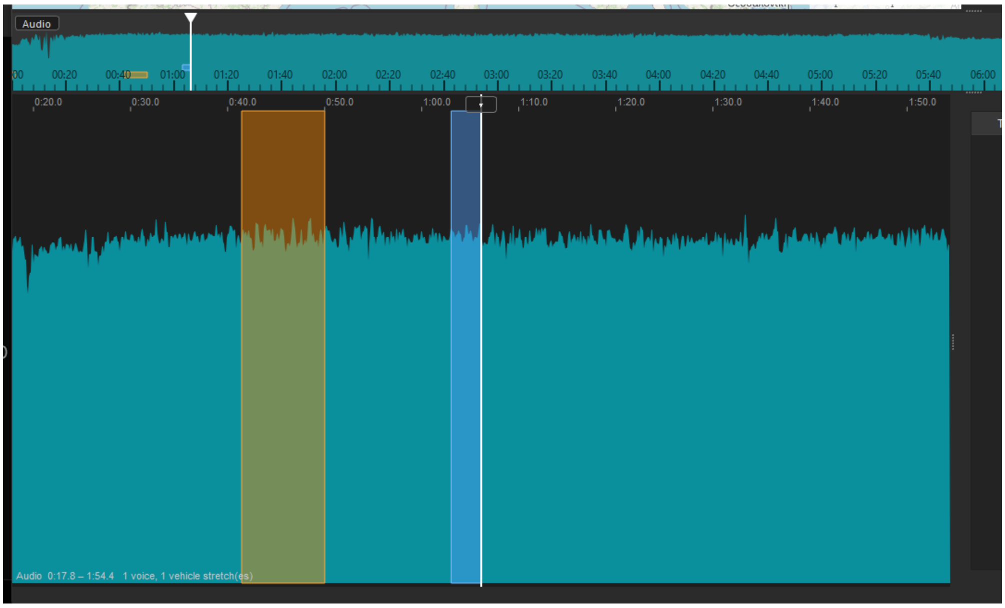
The audio view in the timeline

A small button at the top left of the timeline cycles through **off**, **Video** (the thumbnail strip), **Audio** (the sound level) and **V+A** (the level, half transparent, over the thumbnails). The height of the timeline does not change; the level fills to the bottom edge.

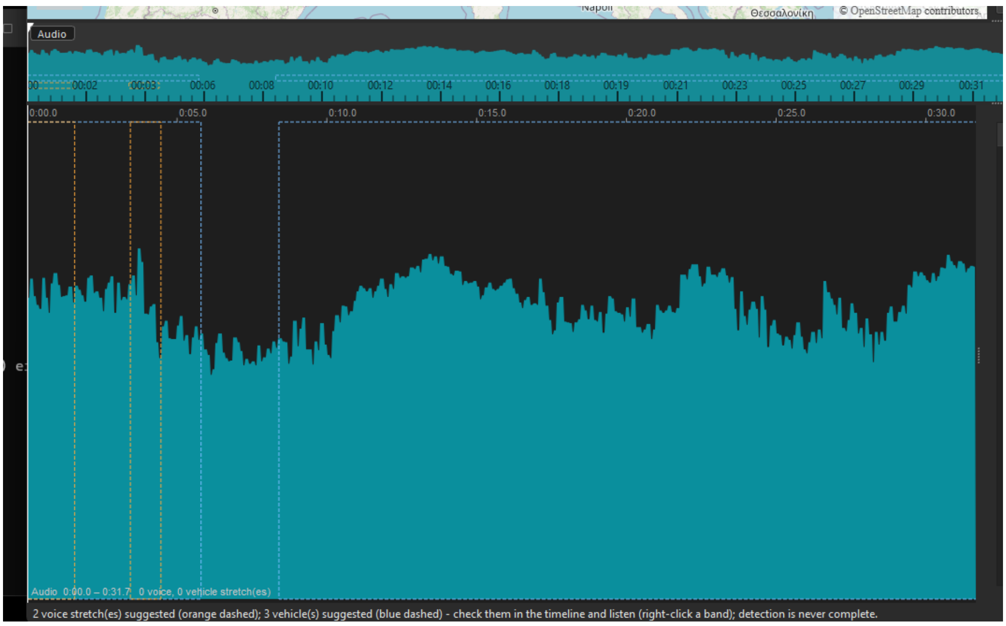
The level comes from the same read as the traffic damper. With **Audio track** on in the Encoder Setup, that read runs in the background as soon as videos are loaded - no dialog, the curve grows file by file, with a line in the status bar and a note in the timeline while it reads (from a slow drive it can take a while). With it off, choosing Audio reads on the spot instead. The audio view works in every build.

The Audio Zoom

A module for any switchable window (like Chart-Flow): the sound level around the playing position, a 60 s window (mouse wheel 8-900 s) that follows the video while the mouse is outside. Marked stretches are orange (voices) and blue (vehicles) bands, the [- and -] marks of the timeline show here too:



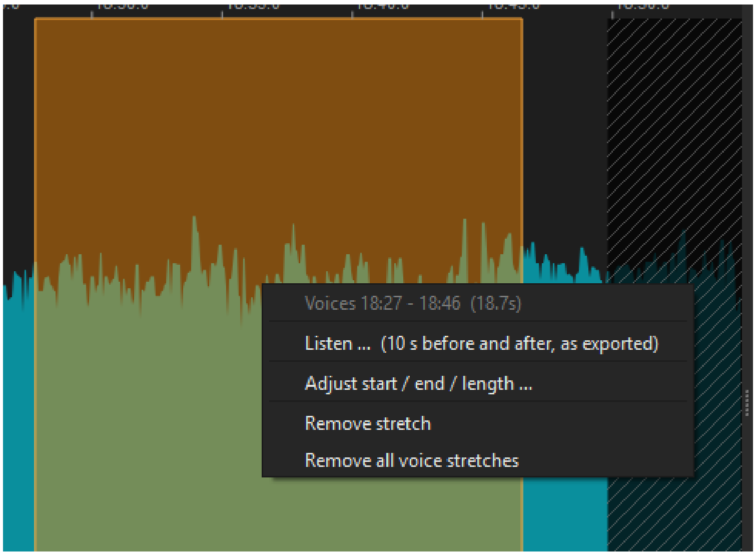
Detect suggestions appear as dashed frames until taken over:



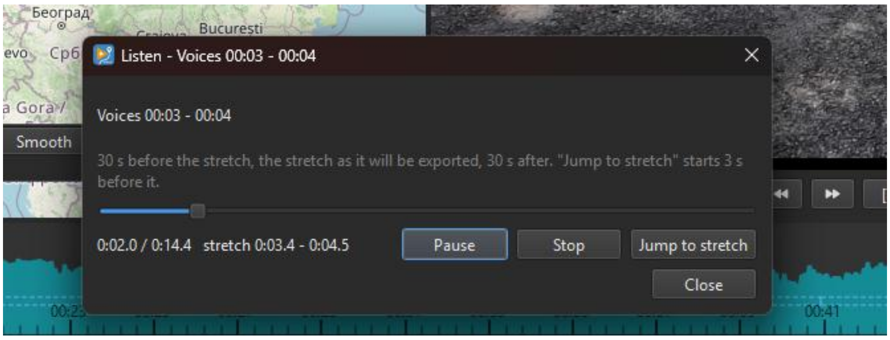
Drag over the curve to mark a new stretch with voices, grab a band's edge or middle to adjust it (see below), right-click a band for its menu, click to jump. Cuts are drawn dark and hatched over it, and what lies under a cut is neither shown nor counted. Add-on only.

Working with a band - Listen, adjust, remove

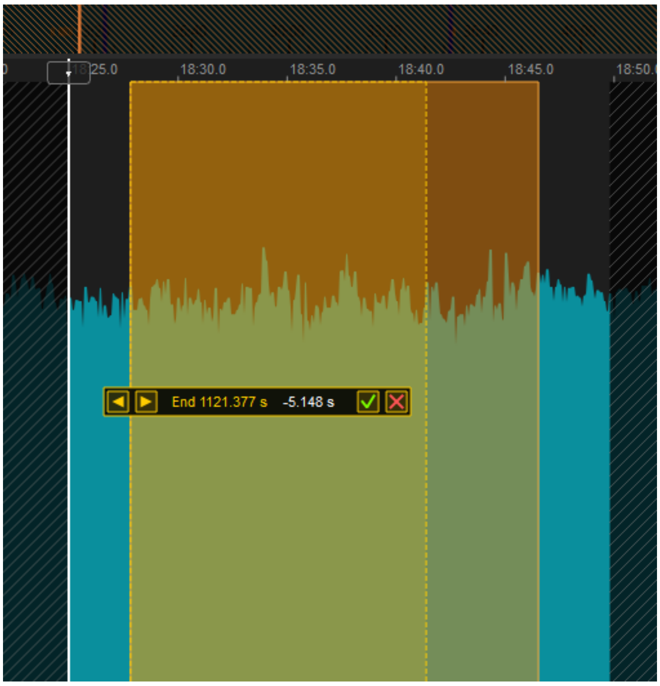
Right-click a solid band. In the timeline the menu offers Listen and Remove; in the Audio Zoom, where the bands are tall enough to grab, it also offers Adjust:



Listen ... renders 10 s before the stretch, the stretch as it will be exported and 10 s after, small and fast, and plays it in a window with Play/Pause, Stop and Jump to stretch (3 s before it).

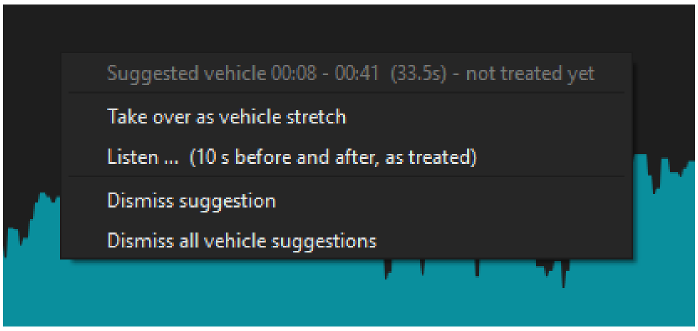


Adjust start / end / length ... opens a fine editor in the Audio Zoom - the same idea as moving or resizing a video cut. Grab an edge to change the start or the end, grab the middle to move the whole stretch, and drag; a small bar then shows the value, with arrows to nudge it and a green check to apply or a red cross to cancel. The arrow keys nudge too (1 ms, with Shift 10 ms, with Alt 100 ms), Enter applies, Esc cancels. You can also grab an edge or the middle of a band directly in the Audio Zoom, without the menu. Adjusting is only in the Audio Zoom, not in the timeline.



Remove stretch takes it away, **Remove all** clears them. A vehicle band also offers **Fill** - where the ride noise comes from: before the stretch (default), after it, or none.

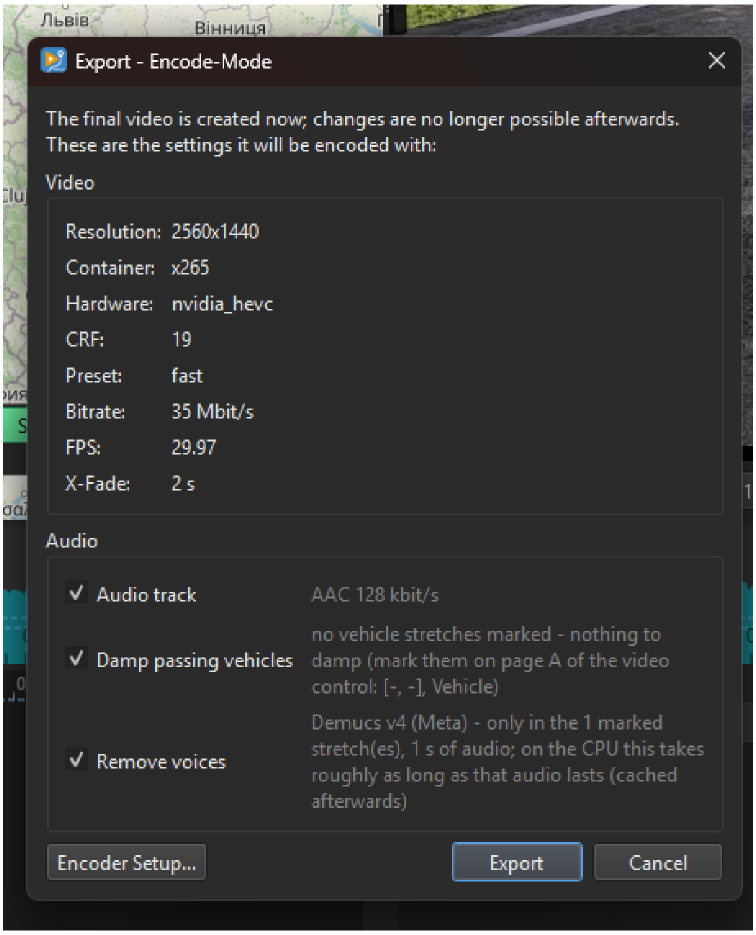
Right-click a **dashed suggestion** instead and it offers to take it over as a real band, to Listen (rendered as it would sound after taking it over), or to dismiss it:



Every change is on the Undo stack. Stretches are stored per file in the project (voice_regions, vehicle_regions); suggestions are not, a fresh Detect brings them back. Without a marked stretch the voice remover separates the whole kept material and the damper does nothing.

The export dialog

Export in Encode-Mode no longer just asks “Are you sure?”. The dialog lists the encoder settings as saved and carries the three sound switches - Audio, Damp passing vehicles, Remove voices - to flip right there, each with a line saying what it will cost:



What you flip here is written to the settings, as OK in the Encoder Setup would; **Encoder Setup…** opens the full setup, **Export** starts the encode.

Command line options

KVRouite can be started with options, one or two dashes, both are accepted (KVRouite.exe -v, KVRouite.exe --verbose). On macOS the program inside the bundle is KVRouite.app/Contents/MacOS/KVRouite.

-? / -h / --help
Prints the list of options and exits.

-v / --verbose
More output on the console. Useful when reporting a problem: start from a command prompt and copy what appears.

-selftest / --selftest
Checks the installation - files, symbols, reading and writing GPX, cutting a video - and exits. The result is printed on the console.

-menuup / --menu-up
Opens the "…" menu of the GPX bar upwards instead of downwards. Meant for screen recordings: at the bottom of the window the menu would otherwise slide over the taskbar. Nothing is stored.

Youtube-Tutorials:

https://www.youtube.com/@KVRouite

email: bernd@kvrouite.com
Web: http://kvrouite.com
Github: https://github.com/ridewithoutstomach/KVRouite

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